OXFORD AQA
INTERNATIONAL QUALIFICATIONS

INTERNATIONAL GCSE
COMPUTER SCIENCE

Paper 1 Programming

November 2024

Preliminary Material

To be opened and issued to candidates 3 months prior to exam, subject to the instructions given in the Teacher's Notes (9210/1/TN).

Note

- The Preliminary Material and Skeleton Program are to be seen by candidates and their teachers **only**, for use during preparation for the November 2024 examination. They **cannot** be used by anyone else for any other purpose, other than that stated in the instructions issued, until after the examination date has passed. They must **not** be provided to third parties.

Information

- A Skeleton Program is provided separately by your teacher and must be read in conjunction with this Preliminary Material.
- You are advised to familiarise yourselves with the Preliminary Material and Skeleton Program before the examination.
- A copy of this Preliminary Material and the Skeleton Program will be made available to you in hard copy and electronically at the start of the examination.
- You must **not** take any copy of the Preliminary Material, Skeleton Program or any other material into the examination room.

INSTRUCTIONS FOR CANDIDATES

Electronic Answer Document

Answers for all questions in all sections must be entered into the word-processed document made available to you at the start of the examination and referred to in the question paper rubrics as the **Electronic Answer Document**.

Preparation for the examination

You should ensure that you are familiar with the **Preliminary Material** and the **Skeleton Program** for your programming language.

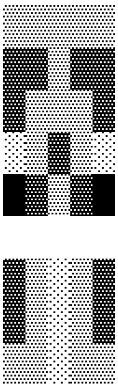
Skeleton program

The skeleton program allows the display of small images.

Each pixel from the image is represented on the screen by using a Unicode character.

\u0020		“whitespace”
\u2591	░	“light shade”
\u2592	▒	“medium shade”
\u2593	▓	“dark shade”
\u2588	█	“full block”

For example, the file `pattern.txt` can be loaded into the skeleton program and will use the Unicode characters to produce the following:



The skeleton program allows the following functionality:

Load image	To load in an image from a text file representation
Display image	To display an image which has been previously loaded
Run length encode image	To display result of run length encoding image
Stretch image	Stretches an image (user can enter a stretch factor)
Lighten image	Alters each pixel in an image to make it lighter
Darken image	Alters each pixel in an image to make it darker

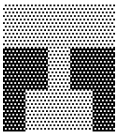
Run length encoding

Run length encoding can be a means of compressing image data. For a large image with lots of repeating pixels, run length encoding can reduce the file size needed to represent the image.

For example, the file `pattern.txt` can be loaded into the skeleton program and it can be passed into the run length encoding routine to produce the following:

```
(5,2)(2,3)(1,2)(3,3)(3,2)(1,3)(1,1)(1,2)(1,3)(1,2)(1,1)(1,4)(1,3)
(1,2)(1,3)(1,4)(5,0)(1,3)(1,2)(1,1)(1,2)(2,3)(1,2)(1,1)(1,2)(1,3)
(2,2)(1,1)(2,2)
```

These values are produced because, if we consider the first 3 rows of the image:



- We start with 5 pixels of the same value (shade 2) `(5,2)`
- We follow with 2 pixels of the same value (shade 3) `(2,3)`
- We follow with 1 pixel (shade 2) `(1,2)`
- We follow with 3 pixels of the same value (shade 3) `(3,3)`
- We follow with 3 pixels of the same value (shade 2) `(3,2)`
- We follow with 1 pixel (shade 3) `(1,3)`

The first value of each pair represents the 'run length' and the second value represents the shade.

END OF PRELIMINARY MATERIAL